Down the straight line that leads you up

X16 (2016) — English translation

An ideal polyatomic gas participates in a process, the states of which are depicted in the figure by points on the straight line $p = p_M(1 - \alpha V)$, where α is a certain constant coefficient, V is the current value of the volume. Molar heat capacity C_V does not depend on temperature.

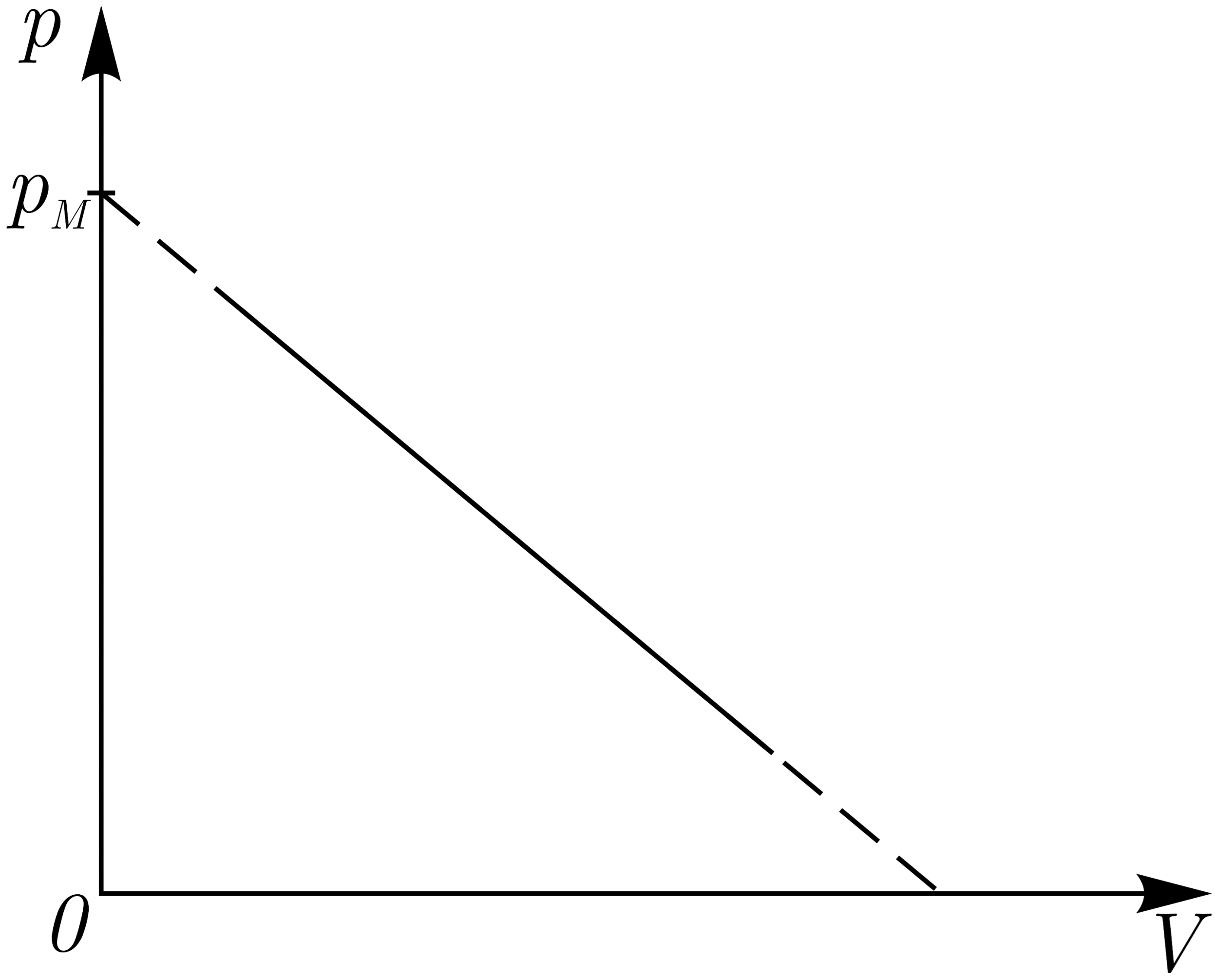

Fig. 1.

A1	At what pressure p_0 does the heat capacity of the gas become zero?	**2.00**

A2 The process involves an ideal Clausius gas, its equation of state is **3.00**

$$p = \frac{RT}{V - b}.$$

Which of these gases has the highest temperature at the point corresponding to the maximum heat capacity? Calculate the temperature difference ΔT for these points. It is known that $b = 44\ \text{cm}^3/\text{моль}$, $R = 8.314\ \text{Дж}/(\text{моль} \cdot \text{К})$, $p_M = 1.51\ \text{МПа}$, $b \ll V_M$, where $V_M = p_M/\alpha$ is some fixed volume.

A3	What value does the molar heat capacity C of an ideal polyatomic gas participating in the process described in paragraph A1 tend to as the volume tends to zero?	**2.00**